90

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CHE 321; Spring – 2020; Quiz – 2   100 Points

1. (15 points) A first order reaction is carried out in 2 equal sized reactors. Select all options that are true about the overall conversion of the reaction:
   a. higher with 2 PFRs in series than with 2 PFRs in parallel. ✓
   b. Changes with the order of PFR and CSTR (PFR-CSTR or CSTR-PFR). ✓
   c. Remain the same for 2 PFRs in series or 2 PFRs in parallel.
   d. lower with two CSTRs in series then with two CSTRs in parallel.

5

2. (15 points) The most unsuitable reactor for carrying out reactions in which high reactant concentration favors high yields is
   a. series of CSTR
   b. one CSTR
   c. plug flow reactor
   d. series of PFRs

15

same vol.

3. (15 points) 'N' plug flow reactors in series with a total volume 'V' gives the same conversion as a single plug flow reactor of volume 'V' for __________ order reactions.
   a. third
   b. first
   c. any
   d. second

15

$k[A]^a$

4. (15 points) The ratio of volume of CSTR to the volume of P.F.R (for identical flow rate, feed composition and conversion) for zero order reaction is
   a. 1
   b. ∞
   c. 0
   d. >1

15

r = k

15

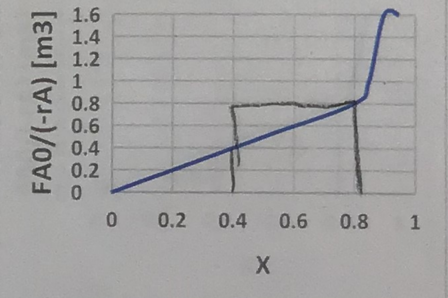
5. (15 points) Explain the answer above

For a zero order rxn, rate is constant, not depend on reactant concentrations. So an integral for PFR would be equal to the area of rectangle as for CSTR. Thus, same volume for same final conversion as they are one line horizontal in levenspiel plot.

answer for 5

~~more volume than PFR for same ... volume ... dependent on final conversion ... more volume. Also, since PFR reacts at the higher possible concentration of A at each point it has a higher rate compared to CSTR which reacts at exit conc. of A, so PFR has more efficiency and thus lower volume for same conversion.~~

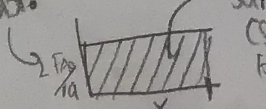
6. (25 points) The following Levenspiel plot shows the response of a chemical reaction. A PFR and then a CSTR in series show a conversion of 0.4 and 0.8 (based on the entry concentration of the PFR). Find the volumes of the PFR and the CSTR.



PFR: .4(.4)(1/2) = .08 m3

CSTR: (.4)(.8) = .32 m3

25



volume same for CSTR & for PFR